

CSA 0378-DATA STRUCTURES

1. Write a C program to perform Matrix Multiplication

```
1 #include<stdio.h>
2 void main()
3 {
4     int a[10][10], b[10][10], c[10][10];
5     int i, j, k, r1, c1, r2, c2;
6
7     printf("Enter rows and columns of matrix A: ");
8     scanf("%d %d", &r1, &c1);
9     printf("Enter rows and columns of matrix B: ");
10    scanf("%d %d", &r2, &c2);
11
12    if(c1 != r2)
13    {
14        printf("Matrix multiplication not possible!");
15        return;
16    }
17
18    printf("Enter elements of matrix A:\n");
19    for(i=0; i<r1; i++)
20        for(j=0; j<c1; j++)
21            scanf("%d", &a[i][j]);
22
23    printf("Enter elements of matrix B:\n");
24    for(i=0; i<r2; i++)
25        for(j=0; j<c2; j++)
26            scanf("%d", &b[i][j]);
27
28    for(i=0; i<r1; i++)
29        for(j=0; j<c2; j++)
30        {
31            c[i][j] = 0;
32            for(k=0; k<c1; k++)
33                c[i][j] += a[i][k] * b[k][j];
34        }
35
36    printf("Result matrix:\n");
37    for(i=0; i<r1; i++)
38    {
39        for(j=0; j<c2; j++)
40            printf("%d\t", c[i][j]);
41        printf("\n");
42    }
43 }
```

Enter rows and columns of matrix A: 2 2
Enter rows and columns of matrix B: 2 2
Enter elements of matrix A:
12 14
11 10
Enter elements of matrix B:
10 11
12 12
Result matrix:
288 300
230 241
=== Code Exited With Errors ===